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Research paper

Collaborative energy-saving path planning of unmanned surface vehicle cluster based on multi-head attention mechanism and multi-agent deep reinforcement learning☆

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ABSTRACT

This paper proposes a Multi-Agent Deep Reinforcement Learning (MADRL) algorithm to address the overlooked issue of energy consumption optimization in existing cluster path planning methods, enabling collaborative energy-efficient path planning for Unmanned Surface Vehicle (USV) clusters. Meanwhile, it provides an alternative MADRL approach to address practical engineering challenges. We not only consider the problems of avoiding danger, reaching the target point and avoiding path conflicts between USV, but also further consider the energy consumption and speed planning of USV cluster. Specifically, firstly, we establish the USV motion model and the USV cluster conflict model, and propose a new energy consumption model, which considers the relationship among speed, marine environment and propulsion load. Secondly, we propose a multi-head local state relevance capture mechanism-multi-agent proximal policy optimization (MLSRC-MAPPO) algorithm. This algorithm can use multi-head attention mechanism (MHA) to capture the potential dependencies between USV, thus enhancing the convergence performance of multi-agent. Finally, in order to reduce the training difficulty, we propose a multi-dimensional action space method for action networks. The experimental results demonstrate that the proposed multi-dimensional action space method has achieved significant success in the lightweighting of the action network: the number of network parameters was reduced by 89.10%, and the computational complexity was decreased by 88.94%, significantly enhancing training efficiency. Meanwhile, the MLSRC-MAPPO algorithm, by incorporating a multi-head attention mechanism, greatly improved the convergence performance of the multi-agent system. In test scenarios with both identical and different starting points, the method reduced energy consumption by 26.24% and 38.47% respectively, fully validating its effectiveness and superiority. Furthermore, comparative experiments with existing cluster energy-saving path planning methods show that the proposed method exhibits clear advantages in terms of energy consumption and path planning efficiency, further verifying its superiority. The corresponding code for this paper is as follows: https://github.com/xhpxiaohaipeng/Multi_USV_Trajectory_energy_plan

1. Introduction

1.1. Research background

With USV's features of high intelligence, flexibility, concealment, low cost, and zero casualties (Aggarwal and Kumar, 2020), it has found extensive applications in tasks such as intelligence investigation, marine exploration, sampling investigation, fire attack, air defense, and anti-submarine operations (Winnefeld and Kendall, 2017). The mission execution capability of a single USV is limited, and it may not be able to successfully complete missions in complex and dynamic ocean environments. However, cooperative mission execution by USV cluster can compensate for the shortcomings of individual USV and fully leverage the advantages of flexible deployment, wide monitoring range, flexible organization, strong resistance to destruction and reconstruction (Eshel, 2023). Nevertheless, efficient collaboration in USV clusters remains a major challenge, with cooperative path planning serving as a core

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component (Sabattini et al., 2017). In multi-agent systems, cooperative path planning requires designing mutually non-interfering trajectories for all cluster members to achieve coordinated movement. The core challenge of USV cluster cooperative path planning lies in establishing an efficient conflict resolution mechanism (Gowalkar and Bhosale, 2023) to ensure spatiotemporal decoupling of multi-agent trajectories and global performance optimization.

As a small-scale intelligent operation platform, USV have limitations in terms of scale, space, and energy storage devices. In water-based tasks, especially in marine exploration, long endurance and low energy consumption are key requirements for USV. Therefore, energy consumption optimization is critical for enhancing USV cluster endurance (Xu et al., 2020). However, current research on USV cluster path planning mostly neglects the issue of energy consumption.

At present, the research on path planning for USV cluster primarily focuses on heading planning and pays less attention to the coordinated planning of speed and heading. However, speed is crucial for the navigation of USV cluster. By adjusting the speed and heading appropriately in path planning, it is possible to improve navigation efficiency and minimize energy consumption to the greatest extent. Consequently, integrating speed considerations into path planning demonstrates substantial research significance and practical value. By considering the coordinated planning of heading and speed, we can further enhance the navigation efficiency, safety, and task execution capabilities of USV cluster. This paper proposes a multi-objective path planning framework for coordinated heading and speed control in USV cluster, with the dual objectives of energy consumption minimization and coordinated energy-saving path planning.

1.2. Related work

Currently, research on path planning for individual USV has reached a relatively mature stage, but several challenges remain in cooperative path planning for USV clusters. The most prominent issue is the potential for path conflicts within the cluster, where multiple USVs may compete for the same spatial position at the same time, leading to collision risks. To address this problem, existing solutions can be broadly categorized into two types: coupled and decoupled methods (Zhang et al., 2020). The coupled approach treats the entire USV cluster as a single system, performing unified planning by integrating the degrees of freedom of all individuals. While computationally efficient, this method overlooks the individual motion characteristics of each USV, thereby limiting the overall maneuverability of the cluster. In contrast, the decoupled approach first generates independent paths for each USV and then resolves conflicts through collision point detection and coordination mechanisms. Although requiring greater computational resources, the methodology effectively reduces path conflict occurrences within the vehicle cluster. Regarding the cluster path planning problem, extensive research has been conducted by numerous scholars. Based on methodological differences, these studies can be roughly classified into traditional cluster path planning methods and deep reinforcement learning (DRL)-based cluster path planning methods (Puente-Castro et al., 2024).

Traditional cluster path planning methods include the artificial potential field method, probabilistic road map method, conflict-based search algorithm, Salp Swarm algorithm, and genetic algorithm. The fundamental objective of multi-agent path planning is to generate collision-free trajectories for all agents within a specified environment, ensuring successful navigation from initial positions to designated targets while satisfying optimization criteria, including total path cost minimization or mission completion time reduction (Stern et al., 2019). Within the domain of multi-agent path planning, Conflict-Based Search (CBS) was proposed in Sharon et al. (2015), featuring a novel two-layer search architecture. The lower layer decomposes the multi-agent path planning problem into independent single-agent planning problems, which are then solved by the A* algorithm. The upper layer performs inter-agent coordination by detecting and resolving path conflicts, thereby enabling collaborative trajectory optimization. However, when map complexity and number of agents increase, CBS's solution time grows significantly. In Boyarski et al. (2015), the authors developed an Improved Conflict-Based Search (ICBS) algorithm that classifies path conflicts into three categories: core conflicts, sub-core conflicts, and non-core conflicts, with optimized priority design for conflict resolution. While maintaining the optimal solution guarantee and completeness of traditional CBS, this improved algorithm employs a multi-valued decision diagram mechanism for conflict classification, albeit with relatively higher computational resource consumption. In Wang et al. (2022a), the authors further improved ICBS, mainly including: first, classifying key conflicts into head-on conflicts and crossing conflicts based on conflict direction; then proposing corresponding conflict resolution strategies accordingly; finally, optimizing the conflict detection method in the high-level search phase. These optimizations substantially decreased the dimensionality of the high-level constraint tree, simultaneously improving search efficiency while significantly lowering computational overhead. In Semiz and Polat (2021), the authors introduced an incremental path planning algorithm at the lower layer of CBS, further reducing computational complexity by optimizing agent path replanning.

For practical applications, in Zhao et al. (2022), the authors proposed a Voronoi diagram-based region partitioning method and an improved Salp Swarm algorithm to solve traditional model predictive control problems, achieving cooperative path planning for USV clusters and effectively addressing search and coverage tasks in aquatic environments. In Wang (2021), the authors utilized a grid method to partition the path planning area for robots and employed the A* algorithm to solve the optimal path problem for multiple mobile robots. In Mohamed et al. (2011), the authors introduced the concept of wall-following in multi-robot path planning and proposed an improved tangent Bug method to overcome local minima issues encountered by the artificial potential field method in real-time path planning, ensuring successful escape from hazardous environments. In Lu et al. (2022), the authors first applied an optimized fuzzy c-means algorithm to divide the search area and then used a hybrid particle swarm optimization algorithm to achieve intra-region path planning, ultimately realizing global path planning for UAV clusters. In Vicmudo et al. (2014), the authors employed a genetic algorithm to solve the cooperative path planning problem for underwater robot clusters, generating the shortest collision-free paths for the cluster. In Lin and Liu (2019), the authors proposed a pheromone-based multi-USV search method for unknown environments. First, the search area was discretized into grids. Then, the pheromone value of each target point was calculated. Next, the target point with the highest pheromone was selected as the next movement goal for the USV. Finally, an artificial potential field-based fitness function was established to enable cooperative rapid search for multiple USVs. In Wei (2023), the authors presented a two-stage cooperative trajectory planning method for quadrotor UAV formations. In the trajectory search stage, the UAV's dynamic characteristics were incorporated into the A* algorithm to discretize the continuous state space into searchable nodes, ensuring the initial trajectory complied with dynamic constraints and reducing computational load for subsequent optimization. In the trajectory optimization stage, polynomial trajectory parameters were solved based on the searched waypoints to obtain smooth and flyable final trajectories. In Fan (2022), the authors addressed the cooperative path planning problem for USV clusters by improving the traditional Beetle Antennae Search algorithm with an ε-greedy strategy to enhance global exploration. They also modified the fitness function for path smoothing and combined it with a genetic algorithm to obtain globally optimal paths even in complex marine environments. After completing the leader's path planning, the global path for the cluster was calculated based on the cluster model. In Li (2021), the authors combined the virtual structure method with the artificial potential field method to achieve cooperative path planning for USV clusters. In Wang (2022), to resolve collision issues in USV cluster path planning, the authors first adopted a global multi-objective path planning method considering ocean currents to generate feasible paths for each USV. Then, a conflict resolution mechanism with priority ordering was introduced to ensure collision-free paths among all USVs, guaranteeing the safety and stability of the cluster.

Traditional path planning methods (such as search algorithms, artificial potential field approaches, swarm intelligence optimization, and bionics-inspired methods) generally exhibit several performance limitations, including suboptimal path quality, excessive computational time, low convergence efficiency, susceptibility to local optima, and insufficient system robustness (Lu et al., 2022). With the advancement of artificial intelligence, many researchers have shifted their focus to RL-based approaches to address these issues. Deep Reinforcement learning (DRL) algorithms demonstrate prominent advantages in USV cluster path planning: (1) Through continuous interaction with the environment and dynamic parameter optimization, they can obtain globally optimal navigation paths while exhibiting excellent environmental adaptability and system stability. (2) The decision-making mechanism avoids complex search operations, making it particularly suitable for real-time path planning in USV clusters. (3) Reward functions based on voyage distance, energy consumption, and other metrics can be flexibly designed to achieve multi-objective cooperative optimization without manual parameter tuning. (4) Independence from prior environmental information enables reliable planning performance even in dynamic scenarios with limited information. For instance, in Zhao et al. (2021), the authors proposed an improved randomized stopping deep reinforcement learning method, enabling USV clusters to achieve high-precision adaptive path tracking. In Nantogma et al. (2023), a hybrid multi-agent deep reinforcement learning framework was designed, not only ensuring obstacle avoidance capability for USV clusters but also significantly improving their autonomous capture performance for maneuvering targets. In Dong and Liu (2021), a UAV cluster cooperative path planning method based on the DDQN algorithm was developed, achieving optimal mission exploration under power capacity and communication constraints through autonomous learning, effectively reducing coverage overlap and flight energy consumption. In Wang et al. (2022b), the authors enhanced the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm, significantly improving target path tracking accuracy for USV clusters. In Liu et al. (2023), a reinforcement learning-based cooperative search method was proposed, enabling USV clusters to maintain efficient collaborative search capabilities even in complex interference environments. Notably, in Wang et al. (2023), the Scalable-MADDPG method overcame the scalability limitations of traditional approaches, achieving coordinated intrusion into target areas by USV clusters within strict time limits. Regarding dynamic environment adaptability, in Wen et al. (2022), the authors developed a MADDPG-based path planning method that endowed USV clusters with dynamic obstacle avoidance and autonomous navigation capabilities. In Xia et al. (2023a), the Multi-Agent Proximal Policy Optimization (MAPPO) algorithm successfully addressed the intelligent encirclement of maritime escape targets, significantly improving both encirclement efficiency and success rates. In Xia et al. (2023b), a MAPPO-based cluster assembly control method was proposed, innovatively solving the cooperative formation assembly challenge for heterogeneous USV clusters. These studies fully demonstrate the unique advantages of reinforcement learning in enhancing the autonomy, adaptability, and coordination of unmanned system clusters.

In the research on multi-agent cooperative energy-efficient path planning, In Li and Yang (2023), the author addressed the challenges of limited robot energy storage and the highly coupled nature of multi-agent path finding (MAPF) by proposing a priority-free ant colony optimization (PFACO) algorithm. This algorithm plans collision-free and energy-efficient paths for multi-robot systems in rugged terrain environments to reduce motion energy consumption. In Shu et al. (2024), on the other hand, the author focused on energy optimization for UAV swarm delivery and introduced an energy-aware multi-agent deep reinforcement learning (EMADRL) algorithm. This approach integrates a UAV energy consumption model into the multi-agent deep reinforcement learning (MADRL) framework to design energy-saving routes for the swarm, thereby reducing overall energy expenditure. In Ao et al. (2023), the author tackled the problem of maximizing communication efficiency in the dynamic deployment of UAV swarms for emergency communication scenarios. The authors proposed a MADRL-based distributed trajectory optimization algorithm—Dual-Stream Attention Multi-Agent Actor-Critic (DSAAC). This algorithm comprehensively considers factors such as UAV throughput, safe distance, and power consumption, effectively reducing the swarm's energy consumption.

In summary, existing research on cluster path planning has primarily focused on various methodologies, including search algorithms, artificial potential field methods, swarm intelligence approaches, bionics-inspired techniques, and reinforcement learning. With the continuous advancement of artificial intelligence, DRL have demonstrated significant technical advantages in the field of cooperative path planning for USV formations. These algorithms effectively address the limitations of traditional cluster path planning methods, such as inadequate performance, prolonged search times, low convergence rates, and susceptibility to local optima. Moreover, DRL exhibits strong adaptability and robustness, making it particularly suitable for dynamic path planning scenarios that require real-time decision-making and multi-objective optimization capabilities. However, it is important to emphasize that none of the aforementioned methods adequately address the cooperative planning of speed and heading for USVs within the cluster, nor do they sufficiently consider energy consumption optimization during navigation. These gaps highlight critical areas for future research to enhance the practicality and efficiency of USV cluster operations. However, although the aforementioned multi-agent cooperative energy-efficient path planning methods aim at minimizing energy consumption, they only consider environmental impacts on energy expenditure while failing to incorporate self-velocity into the energy consumption model. Moreover, all these methods focus solely on path planning without addressing speed planning. To bridge this gap, this paper focuses on the cooperative planning of velocity and heading in clusters while introducing velocity factors into the energy consumption model.

1.3. Motivation and contribution

According to the literature review above, the current issues with USV cluster collaborative path planning methods are as follows:

(1) Currently, traditional USV cluster path planning algorithms require accurate modeling of obstacles in space. When the obstacles in the environment are complex and diverse, it can significantly increase the computational complexity of the path planning algorithm. Especially when the conditions for USV cluster collaborative path planning are complex and involve multiple constraint conditions, traditional methods often face problems of high time complexity and getting trapped in local minima. Additionally, as the demand for USV intelligence increases, traditional USV cluster collaborative path planning algorithms will struggle to meet the requirements for intelligence. In contrast, MADRL has the potential to better meet the demands for intelligence. Therefore, it would have significant academic value and promising application prospects to develop a USV cluster path planning algorithm based on MADRL.

(2) The current cooperative path planning for USV clusters mainly focuses on heading planning while neglecting speed planning, which may result in certain limitations in practical applications.

(3) Existing cooperative path planning for unmanned surface vehicle clusters overlooks the endurance issue and does not consider energy consumption optimization.

(4) In traditional MADRL methods, the dimensionality of an agent's discrete action space is determined by the product of the dimensions of its sub-action spaces (i.e., $a1 \times a2 \times \cdots$). Theoretically, expanding the dimensions of sub-action spaces can enhance the performance of agents and facilitate the planning of smoother paths. However, such expansion can lead to the issue of dimensionality explosion, and this effect becomes increasingly pronounced as the number of sub-action spaces grows. In the application scenario of USV clusters, the phenomenon of dimensionality explosion significantly reduces training efficiency. Specifically, agents are required to handle an exponentially increasing number of action combinations, which not only escalates the computational complexity of training but also leads to challenges such as convergence difficulties (Tan et al., 2019).

(5) The traditional MAPPO algorithm (Yu et al., 2022) typically concatenates local states directly as the global state input to the critic network, which may lead to the loss of potential dependencies among agents and adversely affect training performance.

Therefore, this paper has made the following contributions to address the above-mentioned issues:

(1) We propose an MLSRC-MAPPO algorithm. In the traditional MAPPO algorithm (Yu et al., 2022), it is common to simply concatenate all one-dimensional local states into a larger one-dimensional global state and input it into a critic network to score the policy. However, this approach may lead to the loss of potential dependencies between agents during the training process. In the MLSRC-MAPPO algorithm, first, for the construction of the global state, we do not simply concatenate all one-dimensional local spatial states into a larger one-dimensional global state. Instead, we construct a global state with a two-dimensional structure, preserving the potential dependencies between USV local states. Then, we use the MHA to build a MLSRC model as the critic network of the agent. Finally, we input the global state into the MLSRC to obtain the correlation information between each USV local state. This approach can capture the potential dependencies between each USV through local states, thereby greatly improving the learning ability and coordinated energy-saving path planning capability of the agent.

(2) We propose a new USV energy consumption model by establishing the functional relationship between the propulsion power of the USV and its sailing speed and marine environment. This allows us to reduce energy consumption by planning the USV's sailing speed and heading. During USV navigation, in addition to overcoming water resistance, it is also influenced by factors such as ocean currents, winds, and waves. Therefore, the propulsion power is jointly affected by the sailing speed and marine environment. Current research usually assumes that the USV sails at a constant speed and calculates energy consumption by considering the combined forces of ocean currents, winds, and waves, as well as the sailing distance (Zhang et al., 2022), to establish the USV energy consumption model. In this study, due to the specific functional relationship between the USV propulsion power, sailing speed, and marine environment, we construct an energy consumption model based on the sailing speed and marine environment. In this model, the USV selects the marine environment through its heading and plans its speed to control the propulsion power. The traditional energy consumption model can only plan the course, but our proposed energy consumption model enables coordinated planning of the USV's heading and speed, which better meets the requirements of USV cluster.

(3) We propose a multi-dimensional action space for the agent's action network. Based on discrete acceleration and angular velocity, we adjusted the sailing speed (sub-action 1) and heading (sub-action 2) of the USV, while limiting the maximum speed, heading angle, maximum acceleration, and maximum angular velocity to ensure that the motion planning of the USV complies with kinematic constraints. In our proposed multi-dimensional action space method, different output layers were assigned to different sub-actions in the action network, allowing for separate outputs for each sub-action. By doing so, the action space of the agent is the sum of each sub-action space ($a1 + a2 + \cdots$), effectively minimizing the action space and avoiding the issue of dimension explosion, thus significantly reducing the training difficulty.

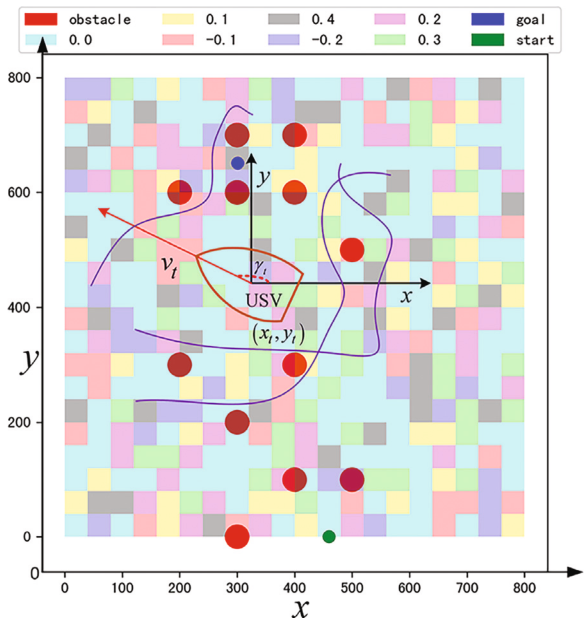

Fig. 1. USV motion model.

(4) We construct a reinforcement learning environment for the USV cluster. Based on the motion and energy consumption model of the USV, as well as the USV cluster conflict model, we defined the state space, action space, and reward function. In order to guide the agent to effectively plan energy-saving paths, we carefully designed the reward function. The reward function comprehensively considers distance, time consumption, energy consumption, obstacle avoidance ability and dynamically restricts the weight of energy consumption. Finally, we verified the effectiveness and superiority of the proposed method through experiments.

2. Establishment of USV model and reinforcement learning environment

To accomplish path and speed planning for USV clusters, we established motion model, energy consumption model, and USV cluster conflict model for the USVs. Taking into account the influence of USV motion and the marine environment, our research is based on the following assumptions: (1) We consider both static hazard areas and dynamic obstacles. (2) It is assumed that oceanic regions are distributed in a grid-like fashion, with consistent marine environmental conditions prevailing within each delineated area. Furthermore, the USV possess the capability to detect marine environmental conditions within a 50 nm radius. (3) Given the relatively slow and small-amplitude variations in vertical position and attitude of the USV, we simplify its motion model to two-dimensional horizontal plane motion, disregarding roll, pitch, and heave motions (Hu et al., 2023). It is crucial to emphasize that these assumptions do not compromise the validity of the verification results obtained for the proposed method in this study.

2.1. Motion model of USV

Fig. 1 illustrates the schematic diagram of the USV's motion model, where distinct sea conditions are represented by differently colored rectangular regions. The hazardous area is marked by a red circular region, while the purple line denotes the movement trajectory of dynamic obstacles. Additionally, the blue and green points indicate the target area and the starting point, respectively. The coordinate adjustment of the USV is given by (1), and the motion range of the USV needs to be strictly limited as indicated by (2). The kinematic constraints of the USV are described by (3) (Hu et al., 2023).

$$\begin{cases} x_{t+1} = x_t + \dot{x}_t \Delta t \\ y_{t+1} = y_t + \dot{y}_t \Delta t \end{cases} \tag{1}$$

$$\begin{cases} x_{min} \le x_t \le x_{max} \\ y_{min} \le y_t \le y_{max} \end{cases} \tag{2}$$

$$\begin{cases} \dot{x}_t = v_t \cos\gamma_t \\ \dot{y}_t = v_t \sin\gamma_t \\ \dot{v}_t = d_t \\ \dot{\gamma}_t = w_t \end{cases} \tag{3}$$

where x_{min} and x_{max} represent the motion boundaries in the x-direction, y_{min} and y_{max} represent the motion boundaries in the y-direction, Δt represents the time interval for USV planning, (x_t, y_t) represents the position coordinates of the USV at time t, v_t represents the velocity of the USV, γ_t represents the heading angle of the USV, d_t represents the acceleration of the USV, and w_t represents the angular velocity of the USV. The adjustment of the speed and heading is given by (4), the speed, heading, angular velocity, and acceleration are subject to constraints as demonstrated in (5).

$$\begin{cases} v_{t+1} = v_t + d_t \Delta t \\ \gamma_{t+1} = \gamma_t + w_t \Delta t \end{cases} \tag{4}$$

$$\begin{cases} 0 \le v_t \le v_{max} \\ 0 \le \gamma_t \le 2\pi \\ w_{min} \le w_t \le w_{max} \\ d_{min} \le d_t \le d_{max} \end{cases} \tag{5}$$

where v_{max} represents the maximum cruising speed of the USV, d_{min} and d_{max} represent the minimum and maximum accelerations of the USV, and w_{min} and w_{max} represent the minimum and maximum angular velocities of the USV.

2.2. Energy consumption mode of USV

The USV is powered by an Energy Storage System (ESS) to provide power support. During navigation in still water, the USV encounters reactive forces and water friction, which must be overcome by its propulsion system. The relationship between the propulsion power of the USV and its velocity in still water is given by (6) (Liang et al., 2022; Hein et al., 2021).

$$P_{pl,t} = c_1 (v_t)^{c_2} \tag{6}$$

where $P_{pl,t}$ denotes the propulsion power of the USV, c_1 and c_2 represent the coefficient of proportion and the coefficient of exponent respectively, c_1 is the static water resistance coefficient, and c_2 is a value only related to the ship's shape. Generally, c_1 and c_2 are fixed values.

In real ocean environments, the USV is influenced by additional resistances generated by sea wind, waves, and currents (Liu et al., 2023; Xia et al., 2023a,b), thus requiring adjustment of the relationship between the propulsion power of the USV and its cruising speed as demonstrated in (7) (Liang et al., 2022; Hein et al., 2021).

$$P_{pl,t} = c_1 (1 + f_s)(v_t)^{c_2} \tag{7}$$

where f_s is the additional resistance coefficient (ARC), which is closely related to changes in the ocean environment. It is usually determined by the magnitude of sea wind, waves, currents, and the relative motion direction with respect to the USV (Liang et al., 2022; Hein et al., 2021). As can be seen, when $f_s > 0$, the ocean environment provides resistance to the USV, when $f_s < 0$, the ocean environment provides assistance. The additional resistance coefficients in different regions of the ocean environment are shown in the rectangular region in Fig. 1. Therefore, the total power of the USV can be expressed as (8), and the energy consumption of the USV is given by (9).

$$P_t = (P_{pl,t} + P_{se}) < P_d^{max} \tag{8}$$

$$E_t = P_t \Delta t \tag{9}$$

where P_{se} represents the power consumed by the USV for communication, navigation, and other service loads, P_t represents the total power of the USV at time t, and P_d^{max} represents the maximum discharge power of the ESS, E_t represents the energy consumption of the USV. Since the power required for steering the USV is relatively small compared to the propulsion power and has little impact on the experimental results, the energy consumption for steering can be ignored.

2.3. USV cluster conflict model

In the path planning algorithm for USV clusters, in addition to avoiding danger, USV also need to solve the problem of path conflicts between each other. During USV cluster path planning, real-time position monitoring of each vehicle is essential to prevent inter-vessel collisions and ensure overall cluster safety. Based on the direction of motion of two conflicting USVs, key vertex conflicts are divided into opposite vertex conflict and intersecting vertex conflict (Gowalkar and Bhosale, 2023; Sharon et al., 2015).

The opposite vertex conflict can be visually demonstrated by the case shown in Fig. 2. An opposite vertex conflict (U_i, U_j, A, t) is defined when U_i and U_j have completely opposing motion directions, resulting in an exchange of their occupied vertices after passing through conflicting vertex A at time step t. In Fig. 2, U_i departs from the starting vertex $A2$ and moves towards the target vertex $A5$, while U_j departs from the starting vertex $A4$ and moves towards the target vertex $A1$. The planned paths for U_i and U_j are $p_i = \{A2, A3, A4, A5\}$ and $p_j = \{A4, A3, A2, A1\}$, respectively. At time step 1, U_i and U_j both occupy vertex $A3$, resulting in conflict $c_1 = \{U_i, U_j, A3, 1\}$. At time step 0, U_i occupies vertex $A2$ and U_j occupies vertex $A4$. At time step 2, U_i occupies vertex $A4$ and U_j occupies vertex $A2$, indicating that they exchange the vertices occupied by each other after the conflict occurs.

The intersecting vertex conflict represents a concept complementary to opposite vertex conflict, distinguished by the fact that U_i and U_j do not exchange their occupied vertices after passing through conflict vertex A. As shown in Fig. 3, the planned paths for U_i and U_j are $p_i = \{B1, B2, B3, B4\}$ and $p_j = \{A2, B2, C2, C3\}$, respectively. U_i and U_j both occupy vertex $B2$ at time step 1, resulting in a vertex conflict $c_2 = \{U_i, U_j, B2, 1\}$. However, they do not exchange the vertices they occupy at time step 2.

2.4. Reinforcement learning environment modeling

To effectively train the agent, adjustments will be made to the heading and velocity of the USV at time intervals of Δt, with a maximum number of adjustments set as T. In the maritime environment, we take into account the constraints on the motion of the USV cluster caused by static obstacles, dynamic obstacles, energy consumption and conflicts between USV clusters. Based on the model established above, we will build a reinforcement learning environment as follows.

2.4.1. Local state space

The agent makes path planning decisions by observing local environmental state changes. The local state space of the USV mainly consists of two parts: the USV's own state and the maritime environmental state. The USV's own state includes position, target point location, speed, heading, energy consumption, and total load power. The maritime environmental state includes additional resistance coefficients at the USV's location, additional resistance coefficients in the surrounding area of the USV, the location of danger zones, and the location of dynamic obstacles. The location of danger zones refer to

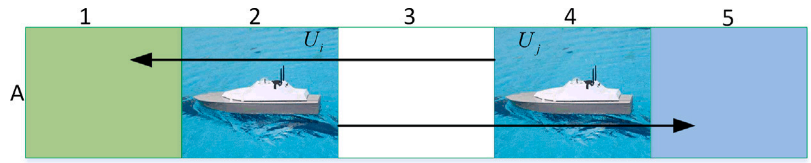

Fig. 2. Example of opposite vertex conflict.

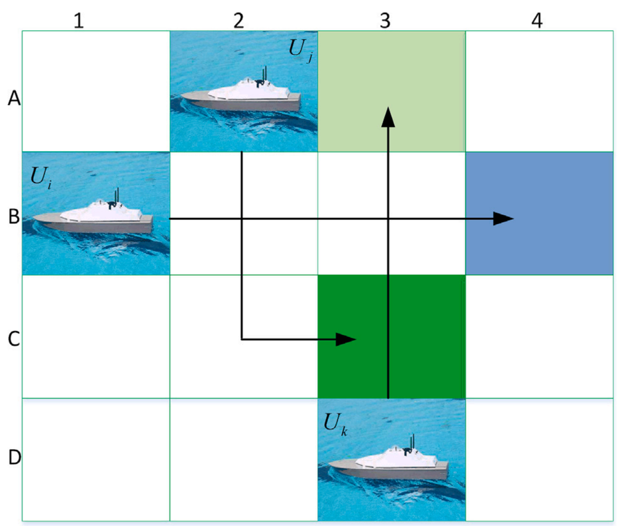

Fig. 3. Example of intersecting vertex conflict.

the coordinates of the centers of the four closest dangerous areas within the detection range of the USV. The location of dynamic obstacles refer to the coordinates of the four closest dynamic obstacles within the detection range of the USV. Therefore, the local state space is expressed as (10).

$$s_t = \{x_t, y_t, x_g, y_g, \dots, E_t, ARC_t, ARC_{L,t}, ARC_{R,t}, ARC_{D,t}, ARC_{U,t}, x_{o1,t}, y_{o1,t}, \dots, y_{o4,t}, x_{do1,t}, y_{do1,t}, \dots, y_{do4,t}\} \tag{10}$$

where ARC_t, $ARC_{L,t}$, $ARC_{R,t}$, $ARC_{D,t}$ and $ARC_{U,t}$ respectively represent the additional resistance coefficients in the USV's position and the left, right, rear, and front marine environment. These coefficients indicate the condition of the maritime environment. x_g and y_g represent the horizontal and vertical coordinates of the center point of the target area. $x_{o1,t}, y_{o1,t}, \dots, y_{o4,t}$ denote the horizontal and vertical coordinates of the centers of the four nearest dangerous areas within the USV's detection range. Similarly, $x_{do1,t}, y_{do1,t}, \dots, y_{do4,t}$ represent the horizontal and vertical coordinates of the four closest dynamic obstacles within the USV's detection range. The value is determined as follows: if the USV detects more than four dangerous areas, then the coordinates of the four closest dangerous areas to the USV, based on their distances, are assigned sequentially to $x_{o1,t}, y_{o1,t}, \dots, y_{o4,t}$. If the number of detected dangerous areas is less than four or no dangerous areas is detected, the corresponding coordinate values are set to $(-x_{max}, -y_{max})$, indicating that no dangerous areas exist at that direction. In the same way, $x_{do1,t}, y_{do1,t}, \dots, y_{do4,t}$ are assigned values by the same method.

2.4.2. Action space

This paper utilizes discretized control commands to achieve motion control of the USV cluster, with the decision output consisting of two-dimensional control parameters: acceleration and angular velocity. The control strategy design involves three critical components: Firstly, uniform discretization is applied to acceleration and angular velocity parameters, converting continuous control quantities into finite action sets. Secondly, to mitigate the curse of dimensionality in the action space, a dimensional decoupling strategy is adopted to construct a hierarchical action space structure (i.e. multi-dimensional action space), where the agent's local control command set can be formally represented by (11). Finally, during command execution, each USV initially updates its velocity and heading parameters according to (4)–(5), followed by iterative position coordinate calculations based on (1)–(3), thereby completing the closed-loop control process for USV cluster motion states.

$$a_t = (d_{min}, d_{min} + \Delta d, d_{min} + 2\Delta d, \dots, (s_d - 1)\Delta d, d_{max}, w_{min}, w_{min} + \Delta w, w_{min} + 2\Delta w, \dots, (s_w - 1)\Delta w, w_{max}) \tag{11}$$

where s_d denotes the size of the sub-action space corresponding to acceleration, Δd represents the absolute difference between adjacent accelerations in the sub-action space, calculated as $(d_{max} - d_{min})/s_d$. Similarly, s_w indicates the size of the sub-action space corresponding to angular velocity, and Δw is the absolute difference between adjacent angular velocities in the sub-action space, calculated as $(w_{max} - w_{min})/s_w$.

2.4.3. Reward function

In the path planning task of the USV cluster, the core goal is to make the USV reach their respective target areas, so we need to calculate the distance between the USV and the center point of the target area in real time, as demonstrated in (12).

$$dis_{g,t} = \sqrt{(x_t - x_g)^2 + (y_t - y_g)^2} \tag{12}$$

where $dis_{g,t}$ represents the distance between the USV and the center point of the target area.

Meanwhile, the USV needs to avoid dangerous areas and dynamic obstacles in the surrounding environment. Therefore, it is necessary to calculate the distance between the USV and the center point of the danger zone as well as the dynamic obstacles in real-time. The specific calculation method is represented as (13).

$$dis_{o/D,n,t} = \sqrt{(x_t - x_{o/D,n})^2 + (y_t - y_{o/D,n})^2} \tag{13}$$

where $dis_{o/D,n,t}$ denotes the distance between the USV and either the centroid of the danger zone or a dynamic obstacle, n represents the nth danger zone or dynamic obstacle. In order to guide the USV to reach the target point while avoiding dangers, penalties and rewards are used to guide the agent. The penalties for danger zones are described as (14)–(15), and the penalties for dynamic obstacles are represented as (16)–(17).

$$PO_{n,t} = \begin{cases} \eta(OT - dis_{o,n,t}) + \tau, if\ dis_{o,n,t} < OT \\ 0, else \end{cases} \tag{14}$$

$$PO_t = \sum_{n=1}^{N} PO_{n,t} \tag{15}$$

$$PD_{n,t} = \begin{cases} \eta(DT - dis_{D,n,t}) + \tau, if\ dis_{D,n,t} < DT \\ 0, else \end{cases} \tag{16}$$

$$PD_t = \sum_{n=1}^{N} PD_{n,t} \tag{17}$$

where $PO_{n,t}$ denotes the penalty for an individual hazardous area, PO_t denotes the total penalty for all hazardous areas, $PD_{n,t}$ denotes the penalty for an individual dynamic obstacle, PD_t denotes the total penalty for all dynamic obstacles, η represents the weight coefficient, τ represents the penalty bias coefficient, OT represents the radius value of the danger zone, and DT represents the danger threshold for dynamic obstacles, all of which are positive numbers.

In order to encourage the USV to reach the target area quickly, appropriate rewards are given to the agent, as demonstrated in (18).

$$RG_t = \begin{cases} \rho - T_g, if\ dis_{g,t} < GT \\ 0, else \end{cases} \tag{18}$$

where RG_t denotes the reward for the USV's arrival at the target area, ρ represents the target reward coefficient, $\rho > T$. T_g represents the number of scheduling times required for the USV to reach the target point, and GT represents the radius value of the target area. The distance calculation formula between USV is expressed as (19).

$$dis_{i,j,t} = \sqrt{(x_{i,t} - x_{j,t})^2 + (y_{i,t} - y_{j,t})^2} \tag{19}$$

where $dis_{i,j,t}$ denotes the distance between two USVs at time step t, $x_{i,t}$ and $y_{i,t}$ represent the coordinates of the ith USV at time step t, $x_{j,t}$ and $y_{j,t}$ represent the coordinates of the jth USV at time step t.

To determine the presence of opposite vertex conflict among USV, we need to assess if each pair of USV occupies each other's positions after movement. This assessment can be done by (20). The punishment for opposite vertex conflict is described as (21)–(22), and the punishment for intersecting vertex conflict is expressed as (23)–(24).

$$\begin{cases} dis_{i,j,(t-1,t)} = \sqrt{(x_{i,t-1} - x_{j,t})^2 + (y_{i,t-1} - y_{j,t})^2} \\ dis_{i,j,(t,t-1)} = \sqrt{(x_{i,t} - x_{j,t-1})^2 + (y_{i,t} - y_{j,t-1})^2} \end{cases} \tag{20}$$

$$PX_{i,j,t} = \begin{cases} \eta(DT - dis_{i,j,(t,t-1)}) + \eta(DT - dis_{i,j,(t-1,t)}) \\ +\tau, if\ dis_{i,j,(t,t-1)}\ and\ dis_{i,j,(t-1,t)} < DT \\ 0, else \end{cases} \tag{21}$$

$$PX_t = \sum_{i=1,j=1}^{(m,m)} PX_{i,j,t} \tag{22}$$

$$PJ_{i,j,t} = \begin{cases} \eta(DT - dis_{i,j,t}) + \tau, if\ \ dis_{i,j,t} < DT \\ 0, else \end{cases} \tag{23}$$

$$PJ_t = \sum_{i=1,j=1}^{(m,m)} PJ_{i,j,t} \tag{24}$$

where $dis_{i,j,(t-1,t)}$ denotes the distance between the position of the ith USV at time step $t-1$ and the position of the jth USV at time step t, $dis_{i,j,(t,t-1)}$ denotes the distance between the position of the ith USV at time step t and the position of the jth USV at time step $t-1$, $PX_{i,j,t}$ denotes the penalty for an individual opposite vertex conflict, PX_t denotes the total penalty for all opposite vertex conflicts, $PJ_{i,j,t}$ denotes the penalty for an individual intersecting vertex conflict, PJ_t denotes the total penalty for all intersecting vertex conflicts, m represents the number of USV, $i \neq j$, (i,j) and (j,i) do not coexist simultaneously.

Without considering energy minimization, the reward function for the agent is expressed as (25).

$$R_t = -(dis_{g,t} + PO_t + PD_t + PJ_t + PX_t) + RG_t \tag{25}$$

where R_t denotes the agent's final reward.

In order to guide the agent towards energy minimization, the energy consumption of the USV needs to be penalized. During the process of approaching the target area, $dis_{g,t}$ will gradually decrease and approach zero, while E_t does not decrease as the target area is approached. Direct incorporation of E_t into the reward function would disrupt the agent's learning process, thereby impeding the USV's ability to reach the target area and accomplish the task. Consequently, this study proposes a dynamic scaling factor K_t to regulate E_t, yielding the final reward function as formulated in (26).

$$R_t = -(dis_{g,t} + PO_t + PD_t + PJ_t + PX_t + K_t E_t) + RG_t \tag{26}$$

where K_t represents the dynamic scaling factor, which is dependent on the distance between the USV and the target area. The adjustment of its value is determined according to (27).

$$K_t = \begin{cases} dis_{g,t}/dis_{g,0}, if\ dis_{g,t}/dis_{g,0} < \psi \\ \psi, else \end{cases} \tag{27}$$

where $dis_{g,0}$ represents the distance between the USV and the target area at the initial time step. The termination condition for training is described as (28), the total reward value is expressed as (29), and the total energy consumption is represented as (30).

$$e_t = \begin{cases} True, if\ dis_{g,t} < GT\ or\ t = T \\ False, else \end{cases} \tag{28}$$

$$R = \sum_{t=0}^{T_e} R_t \tag{29}$$

$$E = \sum_{t=0}^{T_e} E_t \tag{30}$$

where e_t denotes the termination state of training, T_e denotes the number of adjustment steps at training termination, R represents the agent's total reward, E represents the USV's total energy consumption.

3. The cooperative path planning method of USV cluster

3.1. MLSRC-MAPPO algorithm

The MLSRC-MAPPO algorithm adopts an Actor-Critic network structure and utilizes a centralized training and decentralized execution framework. It trains two independent neural networks: an Actor network with parameters θ and a Critic network with parameters ϕ. During the training process, the Critic network calculates the central value function based on the global state, while the Actor network selects actions based on the local state and the central value function from the Critic network to maximize the rewards. After training, the Actor network can select optimal actions based on the local state.

The interactive training process between the agent and the environment in the MLSRC-MAPPO algorithm is shown in Fig. 4. In Fig. 4, s_t represents the combination of local states detected by m USVs, V_t represents the combination of value functions returned by m Critic networks, a_t represents the combination of actions taken by m USVs, and gs_t represents the global state. In this paper, the Actor and Critic networks are not shared among USVs. The Actor network is denoted as π_θ, and it is trained to maximize the objective. The loss function of the Actor network is given by (31)–(32).

$$L(\theta) = \frac{1}{Bm}\sum_{i=1}^{B}\sum_{k=1}^{m}[e[\pi_\theta(s_i^k)]\alpha + \min(H_i^k A_i^k, clip(H_i^k, 1-\varepsilon, 1+\varepsilon))] \tag{31}$$

$$H_i^k = \frac{\pi_\theta(a_i^k|s_i^k)}{\pi_{\theta_{old}}(a_i^k|s_i^k)} \tag{32}$$

where $L(\theta)$ denotes the action network loss, H_i^k represents the probability ratio between new and old policies for the action-state pair (a_i^k, s_i^k), A_i^k denotes the GAE advantage function, B represents the batch training size, m denotes the number of agents, $e[\pi_\theta(s_i^k)]$ represents the entropy of the policy at state s_i^k, α denotes the entropy-controlling hyperparameter, $\pi_\theta(a_i^k|s_i^k)$ denotes the probability of the new policy selecting action a_i^k at state s_i^k, $\pi_{\theta_{old}}(a_i^k|s_i^k)$ denotes the probability of the old policy selecting action a_i^k at state s_i^k.

The Critic network is denoted as V_ϕ, then the loss function of the Critic network is given by (33).

$$L(\phi) = \frac{1}{Bm}\sum_{i=1}^{B}\sum_{k=1}^{m}\max[(V_\phi(gs_i^k) - R_i)^2, \\ (clip(V_\phi(gs_i^k), V_{\phi_{old}}(gs_i^k) - \epsilon, V_{\phi_{old}}(gs_i^k) + \epsilon) - R_i)^2] \tag{33}$$

where $L(\phi)$ denotes the critic network loss, $V_\phi(gs_i^k)$ represents the value estimate of state gs_i^k by the updated Critic, $V_{\phi_{old}}(gs_i^k)$ represents

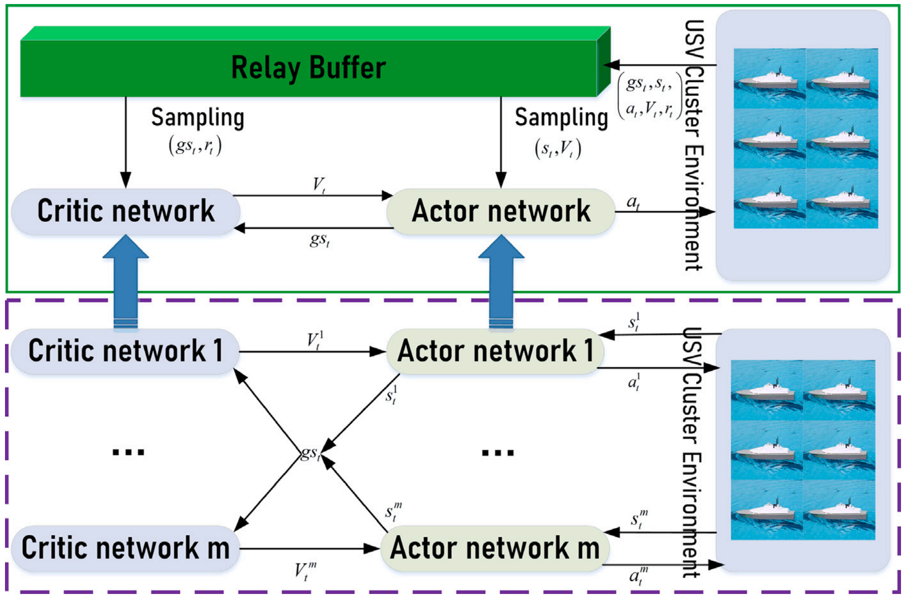

Fig. 4. The training process of the MLSRC-MAPPO agent.

the value estimate of the same state gs_i^k by the old Critic, ϵ denotes the allowable range for value estimate variation, R_i represents the discounted reward.

In conventional MAPPO critic networks, the one-dimensional local states of all agents are typically concatenated into a one-dimensional global state, which is then fed into the critic network for policy evaluation. However, this approach fails to effectively capture and utilize potential dependencies among agents during training. The MHA mechanism has demonstrated remarkable capability in identifying temporal correlations and has been successfully applied across various domains. As evidenced in the literature: Li et al. (2025) and Li et al. (2023) employed MHA for micro-action recognition and body behavior identification respectively, while (Wu et al., 2025, 2023, 2024) applied MHA to video restoration tasks, all achieving significant performance improvements. To address this limitation in capturing latent dependencies, we developed the MLSRC critic network based on MHA. This architecture is specifically designed to effectively model interdependencies among agent states. By processing all agents' local states through MLSRC, we generate a more comprehensive and accurate global state representation. This enhanced representation not only preserves the original state information but also incorporates discovered inter-agent dependencies, thereby providing richer information for policy evaluation during training. Through our proposed MLSRC-MAPPO algorithm, agents can effectively identify latent dependencies among individual USVs within the swarm. This advancement yields dual benefits: it not only enhances agent training efficiency but also significantly improves the cluster's collaborative energy-efficient path planning performance. The specific implementation process is illustrated in Fig. 5.

In the MLSRC-MAPPO algorithm, we preserve the integrity of the local states of different agents. The input global state of the Critic network is represented by (34).

$$gs_t = [s_t^1, s_t^2, \dots, s_t^m] \tag{34}$$

where gs_t denotes the merged global state and is a two-dimensional vector, s_t^m represents the local state of the mth USV, which is a one-dimensional vector. The self-attention mechanism is the foundation for implementing MHA. The process of implementing the self-attention mechanism is described as (35), which is mainly used to capture the relevance between different vectors in a sequence and thus determine the dependency relationship between different USVs.

$$A(Q,K,V) = softmax(\frac{QK^T}{\sqrt{d_{in}}})V \tag{35}$$

where d_{in} represents the size of the local state space for USV, Q, K, and V represent query, key, and value respectively. MHA allows the model to focus on subspace information from different positions, which can more comprehensively capture the correlation between USVs and find potential dependency relationships between USVs. The implementation process of MHA is expressed as (36).

$$\begin{aligned} &MHA(x) = (h_1 \cup \cdots \cup h_h)W_o \\ &where\ h_i = A(x_i W_i^Q, x_i W_i^K, x_i W_i^V) \end{aligned} \tag{36}$$

where h denotes the number of attention heads, x_i represents the ith input segment, h_i denotes the output of the ith attention head, W_i^Q, W_i^K, W_i^V represent the query, key, and value projection matrices for the ith attention head, W_o denotes the output projection matrix after multi-head merging.

The input gs_t is fed into MHA to obtain the relationship between USVs, thereby obtaining a new global state space as demonstrated in (37).

$$gs_t^{MHA} = MHA(gs_t) \tag{37}$$

where gs_t^{MHA} denotes the post-MHA-processed global state embedding interdependencies among USVs.

Subsequently, both gs_t^{MHA} and gs_t are flattened separately, then concatenated and fed into the fully connected layer, as demonstrated in (38)–(40). This design maintains information integrity while incorporating dependencies between USVs into the global state space.

$$gs_t^{new} = f(gs_t) \cup f(gs_t^{MHA}) \tag{38}$$

$$hx = \tanh(\ln(L(gs_t^{new}))) \tag{39}$$

$$value = L(hx) \tag{40}$$

where gs_t^{new} denotes the merged global state, ln represents the layer normalization operation, f indicates the flattening operation, L stands for the linear output layer, and hx signifies the hidden layer output.

3.2. Multi-dimensional action space

When agents adopt a continuous action space, their exploration space becomes extremely large, which significantly increases the difficulty of training in multi-agent tasks. To address this, this paper opts

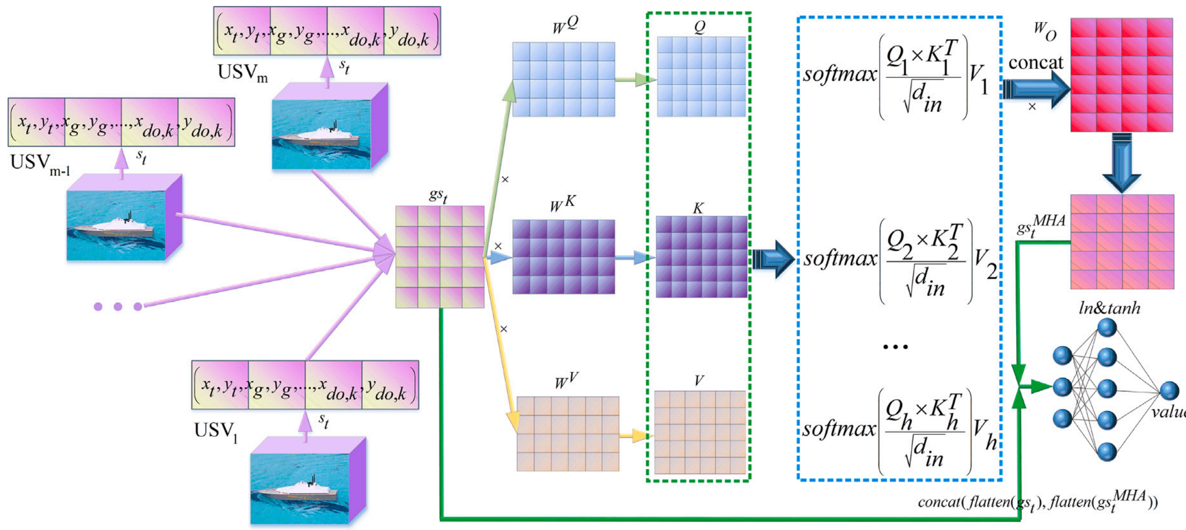

Fig. 5. MLSRC model implementation process.

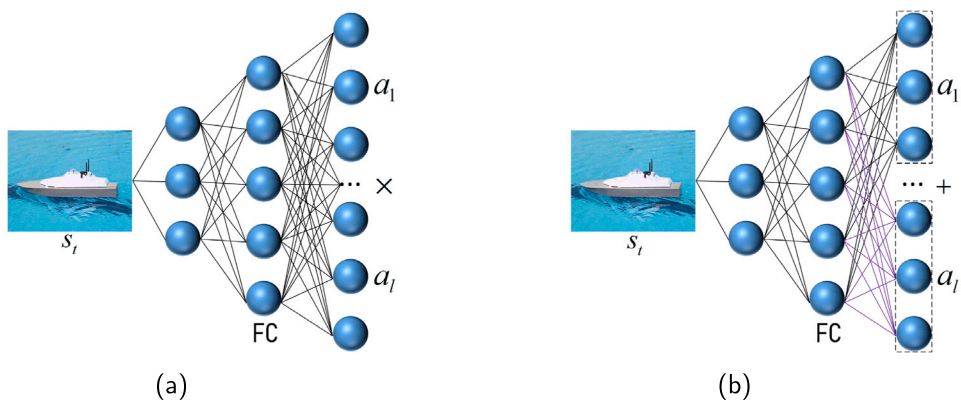

Fig. 6. Actor network (a) Actor network based on traditional action space (b) Actor network based on multi-dimensional action space.

for a discrete action space. However, the size of the traditional discrete action space is determined by the product of the sizes of its sub-action spaces (i.e., $a1 \times a2 \times \cdots$). Typically, a larger sub-action space may lead to better agent performance and smoother path planning, but it can also trigger the issue of dimensionality explosion, especially when the number of sub-action spaces is large. In the context of USV clusters, the impact of dimensionality explosion on training efficiency is particularly pronounced, as agents need to handle a vast number of action combinations, thereby drastically increasing the difficulty and complexity of training. Based on this, this paper proposes a multi-dimensional action space method.

Fig. 6 reveals the fundamental differences between the two action space design paradigms through architectural comparison. The implementation process of the traditional action space is shown in Fig. 6(a), while the implementation process of the multi-dimensional action space is illustrated in Fig. 6(b). The innovative architecture of the multi-dimensional action space, as shown in Fig. 6(b), adopts a hierarchical output design. By decoupling the control strategy, it sets up independent output layers for different sub-actions, enabling the distributed generation of control commands for the entire USV cluster. As a concrete example, we analyze a system comprising two sub-action dimensions, where the size parameter of sub-action 1 is f and that of sub-action 2 is z. The MLP actor network includes one input layer, one hidden layer, and one output layer. When the input layer dimension is ϕ and the hidden layer dimension is h, the total action space size of the traditional action space (Fig. 6(a)) is $f \times z$, and its computational complexity, based on theoretical analysis and calculation, is $2h\phi + 2h(f \times z)$. In contrast, the multi-dimensional decoupled architecture proposed in this paper (Fig. 6(b)) reduces the total action space to $f + z$ dimensions through dimensionality decomposition, and the computational complexity is reduced to $2h\phi + 2h(f + z)$.

Theoretical analysis and computational complexity assessment demonstrate that the conventional centralized action-space architecture suffers from exponential complexity growth when both the quantity of sub-actions and their discretization precision increase concurrently, consequently inducing the dimensionality curse. Notably, in multi-agent collaborative optimization scenarios, this nonlinear surge in computational complexity severely restricts the convergence speed and training efficiency of distributed strategies. In contrast, the multi-dimensional decoupled architecture proposed in this paper maintains a linear growth trend in computational complexity through dimensionality decomposition, significantly mitigating the marginal impact of the number of sub-actions and their discretization level on computational complexity. Theoretical analysis demonstrates that this solution can significantly enhance training efficiency, providing a viable technical pathway for the engineering deployment of large-scale multi-agent systems.

4. Experimental analysis

This paper designs three sets of comparative experiments to validate the algorithm's performance: (1) The first set is the non-energy-constrained baseline group (Non-energy-constrained Minimum Policy

Optimization, NEMP), which employs the MAPPO algorithm for training without considering energy consumption. (2) The second set is the energy-constrained comparison group (Constrained Energy Minimum Policy Optimization, CEMP), which uses the MAPPO algorithm for training while incorporating energy consumption constraints. (3) The third set is the energy-constrained optimization experimental group (Constrained Energy with Relevance Capture, CERC), which applies the proposed MLSRC-MAPPO algorithm for training under energy consumption constraints. By comparing the NEMP and CEMP methods, the effectiveness of the proposed energy consumption model can be verified. By comparing the CEMP and CERC methods, the superiority of the proposed MLSRC-MAPPO algorithm over the original MAPPO algorithm can be demonstrated.

4.1. Experimental environment settings

We study the case of a cluster containing four USVs performing maritime navigation tasks in a fixed sea area of 800 nm×800 nm. In this task, the USV cluster will depart randomly from a certain location and then individually reach their respective fixed target areas. Several fixed dangerous areas exist in this sea area, and the additional resistance f_s of the marine environment to the USVs is divided into different rectangular regions. The parameters for the USV experimental environment are set as follows: $x_{min} = 0, x_{max} = 800$ nm, $y_{min} = 0, y_{max} = 800$ nm, $v_{max} = 40$ kn, $d_{min} = -12$ kn/Δt, $d_{max} = 12$ kn/Δt, $w_{min} = -0.6\pi/\Delta$t, $w_{max} = 0.6\pi/\Delta$t, $P_d^{max} = 60$ kw, $c_1 = 0.0000257, c_2 = 2, P_{se} = 3$ kw, $\Delta t = 20$ min. In this environment, there are a total of 12 circular dangerous areas with a radius of 20 nm and 4 dynamic obstacles. The USV must avoid dangerous areas and dynamic obstacles to reach the target points smoothly. The target areas are circular regions with a radius of 1 nm. The coordinates of the centers of the target areas for USV_1, USV_2, USV_3, and USV_4 are respectively (150, 420), (400, 520), (500, 510), and (200, 380). Rectangular regions with additional resistance values f_s of −0.1, −0.2, 0, 0.1, 0.2, 0.3 and 0.4 (the size is 40 nm × 40 nm) are randomly generated according to the probabilities of 0.1, 0.1, 0.4, 0.1, 0.1 and 0.1, respectively. The departure area for the USV cluster is $x \in [0, 800], y \in [0, 300]$, and it is required that all USVs must safely reach their respective target areas within 32 h.

4.2. Training parameter settings

In the model training phase, we implement randomized initialization by stochastically distributing the USV cluster's starting positions within a predetermined deployment region. The starting points of the USVs may be the same or different to enhance the model's generalization capability. The training process utilizes a large-scale parallel computing architecture, with 512 parallel threads set up and a total of $3 * 10^5$ interaction iterations. To evaluate the model's performance in real-time, after every 10 training rounds, 10 fixed starting points are used for validation, and the average validation reward value is calculated to monitor the training progress. Key training parameters are set as follows: training batch size of 128, learning rate of 0.001, maximum adjustment iterations $T = 96, \rho = 240, \psi = 0.6, \eta = 20, \tau = 6$. The action space includes two discrete sub-action spaces for heading and speed, with both the heading control sub-space and speed adjustment sub-space sizes set to 41 (i.e $s_w = 41, s_d = 41$), and the state space size set to 30. To investigate the impact of the number of attention heads (h) on the performance of the CERC method, this paper analyzes its effect on the training efficiency of the MLSRC-MAPPO algorithm through experimental results under different h. The reward variation is illustrated in Fig. 7. The results show that when $h = 5$, the MLSRC-MAPPO algorithm achieves high convergence speed and reward, when $h = 10$, although the convergence speed decreases slightly, the convergence reward improves significantly compared to $h = 5$. However, when $h > 10$, the convergence reward drops substantially. These findings indicate that $h = 10$ is the optimal setting

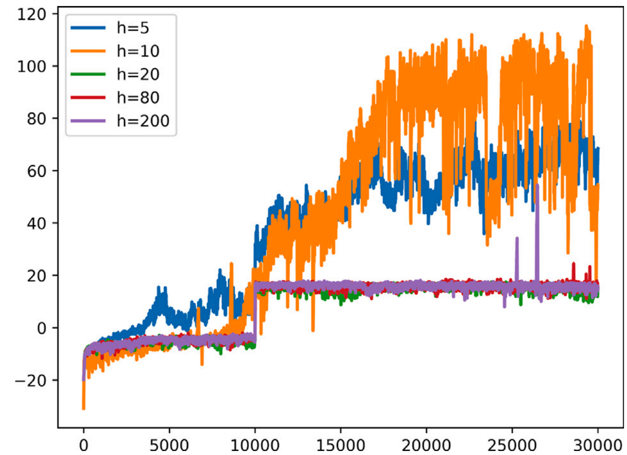

Fig. 7. The trend of reward changes under different numbers of attention heads.

Table 1
The Parameters and FLOPs of different action spaces.

Method	Parameters/KB	FLOPs/K
Traditional action space	116.65	230.13
Multidimensional action space	12.71	25.45

in this experimental environment, and thus h is fixed at 10 for the experiments. After the model training is completed, specific locations within the departure area are selected as the cluster's starting points to comprehensively evaluate the performance of each method.

4.3. Comparison and analysis of training effects

4.3.1. Comparison and analysis of training efficiency between multidimensional action space and traditional action space methods

To validate the optimization effect of the proposed multi-dimensional action space on training efficiency, a quantitative comparative analysis was conducted between the traditional action space and the multi-dimensional action space architectures before training. By calculating two key metrics—the number of parameters and the Floating Point Operations (FLOPs) of the action network—the training difficulty of the agents can be effectively reflected. The specific values are shown in Table 1. The calculation results indicate that, compared to the traditional action space, the multi-dimensional action space architecture achieves an 89.10% reduction in the number of parameters and an 88.94% reduction in FLOPs. This significant dual optimization effect not only verifies the superiority of the multi-dimensional action space method in improving agent training efficiency but also highlights its special value in multi-agent scenarios. Given that multi-agent systems have higher computational complexity and training difficulty compared to single-agent systems, the advantages of the proposed multi-dimensional action space method in reducing computational overhead and enhancing training efficiency are particularly important, fully demonstrating the necessity and urgency of its application. The analysis results also confirm the effectiveness and superiority of the proposed multi-dimensional action space method.

4.3.2. Comparison and analysis of training effects between MAPPO and MLSRC-MAPPO algorithms

Under the constraint of energy consumption minimization, the MAPPO algorithm (i.e., CEMP) and the MLSRC-MAPPO algorithm (i.e., CERC) were used to train the agents, respectively. Both algorithms maintained identical conditions in terms of the training environment and hyperparameter settings. The changes in validation rewards (total

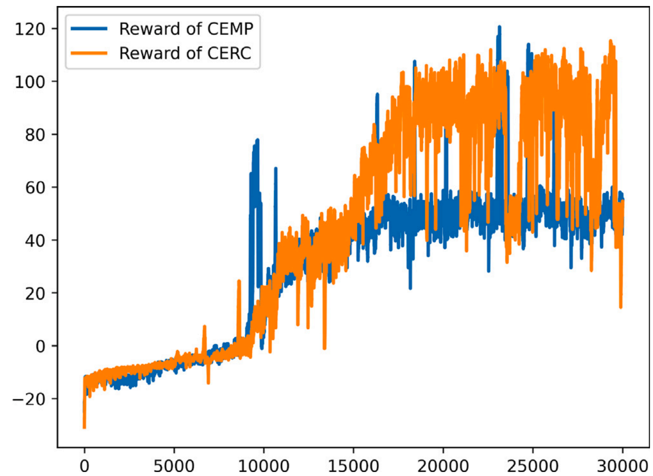

Fig. 8. Training process.

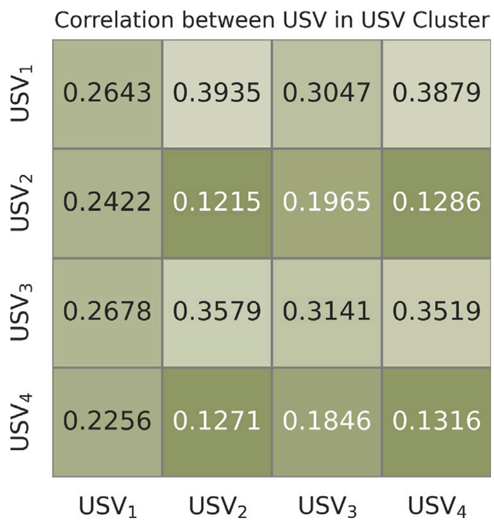

Fig. 9. The attention score of Critic network during training (MLSRC model).

rewards for four agents) during the training process are shown in Fig. 8. The experimental results demonstrate that the CERC method exhibits significantly better convergence performance relative to the CEMP approach. This is primarily attributed to the multi-head attention mechanism in the MLSRC-MAPPO algorithm, which effectively captures the correlations among the USVs in the cluster. By capturing the latent dependencies among the agents, the convergence rewards of the agents are significantly improved, fully validating the superiority of the proposed MLSRC-MAPPO algorithm in terms of performance and convergence characteristics.

To deeply analyze the working mechanism of the multi-head attention mechanism in the MLSRC-MAPPO algorithm, a visualization analysis was conducted on the multi-head attention score matrix of its Critic network (i.e., the MLSRC model). The visualization results in Fig. 9 demonstrate that the MLSRC model not only accurately identifies the dynamic correlation features among different USVs but also deeply reveals the topological dependencies among the units within the cluster system. This capability of association modeling enables the Critic network to more precisely guide the actor network for strategy optimization, ultimately leading to agents based on the MLSRC-MAPPO algorithm exhibiting superior convergence performance compared to those based on the MAPPO algorithm.

Table 2
Detailed planning results of different methods under the same starting point.

Method	No./SP	EC/KWh	ST/h	SD/nm	AS/kn	AA
NEMP	1 (400, 0)	603.53	18.67	500.87	26.83	0.06964
	2 (400, 0)	692.48	15.33	549.47	35.83	0.16522
	3 (400, 0)	676.55	21.33	557.60	26.14	0.03594
	4 (400, 0)	537.93	12.67	468.67	37.00	0.08947
	overall/Average	2510.50	17.00	2076.60	31.45	0.09007
CEMP	1 (400, 0)	518.44	27.00	523.60	19.39	0.04691
	2 (400, 0)	594.24	28.33	548.80	20.64	0.06235
	3 (400, 0)	510.83	27.00	577.67	21.40	0.03827
	4 (400, 0)	509.78	26.67	501.40	18.80	0.14250
	overall/Average	2133.29	27.25	2187.47	20.06	0.07251
CERC	1 (400, 0)	564.51	14.00	496.20	35.44	0.06667
	2 (400, 0)	450.38	28.67	552.20	19.26	0.06512
	3 (400, 0)	455.23	27.00	557.07	20.63	0.03827
	4 (400, 0)	381.55	28.33	494.20	17.44	0.10824
	overall/Average	1851.68	24.50	2099.67	23.19	0.06957

4.4. Test results in the same starting point scenario

After completing the model training, the trained agents underwent performance testing to evaluate their performance in the USV cluster collaborative energy-efficient path planning task. The USVs in the cluster all started from the coordinate (400, 0.0).

4.4.1. Cluster collaborative path planning trajectory analysis

Fig. 10 illustrates the overall effect of the collaborative path planning, where the black lines represent the movement trajectories of dynamic obstacles. The experimental results in Fig. 10 demonstrate that all methods successfully achieved collaborative energy-efficient path planning for the USV cluster, including critical tasks such as hazardous area avoidance, dynamic obstacle evasion, and collision prevention within the cluster, ultimately reaching the target area. Specifically, while the NEMP method chose the shortest path, it did not incorporate environmental factors into its decision-making process. In contrast, the CEMP method, by introducing the proposed energy consumption model, prioritized safer regions with lower ARC values during path planning, validating the effectiveness of the energy consumption model. In-depth analysis demonstrates that the CERC method accomplishes dual optimization relative to CEMP: it simultaneously identifies regions with reduced ARC values while substantially decreasing navigation distance. This stems from the dynamic coupling model among agents established by the multi-head attention mechanism in the MLSRC-MAPPO algorithm. By capturing the spatiotemporal correlation features of the cluster, the CERC method demonstrates significant advantages in path optimization and energy consumption control, fully showcasing the technical advancement of the algorithm in complex cluster collaborative tasks.

4.4.2. Precision analysis of cluster collaborative path planning results

Table 2 provides detailed performance metrics for path planning, including the starting point (SP), sailing time (ST), sailing distance (SD), energy consumption (EC), average speed (AS), and average ARC (AA). The experimental results indicate that the USV cluster successfully completed the mission of reaching the target area within 32 h. Specifically, compared to the NEMP baseline method, the CEMP method achieved a 15.03% reduction in total energy consumption, with a significant decrease in energy consumption per vessel. This optimization stems from the CEMP method's strategy of reducing sailing speed (an average decrease of 36.22% compared to NEMP) and selecting regions with lower ARC values (an average ARC reduction of 19.50%), fully validating the effectiveness of the proposed energy consumption model.

Extended analysis demonstrates that the CERC method achieves significantly enhanced energy efficiency: it reduces total energy consumption by 26.24% compared to NEMP, and representing an 11.21%

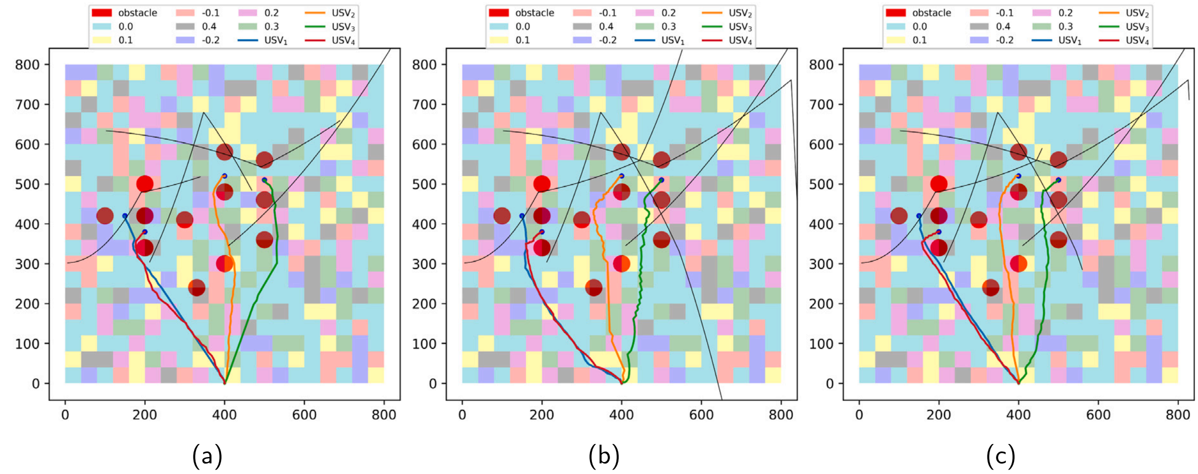

Fig. 10. Collaborative planning results of different methods, (a) NEMP (b) CEMP (c) CERC.

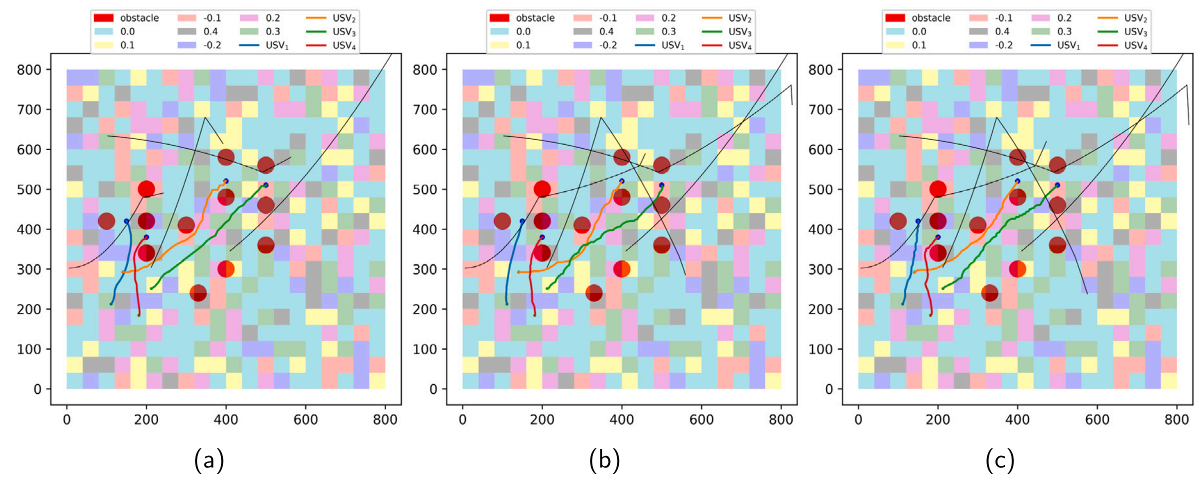

Fig. 11. Collaborative planning results of different methods, (a) NEMP (b) CEMP (c) CERC.

improvement over CEMP. Relative to the CEMP method, the CERC method better integrates considerations of sailing speed, sailing distance, and the relationship between ARC and energy consumption, resulting in the lowest total energy consumption for the cluster. This significant advancement stems from the MLSRC-MAPPO algorithm's innovative incorporation of a multi-head attention mechanism, which enables globally optimal energy-efficient path planning through effective extraction of spatiotemporal correlation features within the cluster. This ensures task completion while minimizing the cluster's total energy consumption. This series of experimental results fully confirms the superiority of the MLSRC-MAPPO over traditional MAPPO in solving complex constraint problems.

4.5. Test results in the different starting point scenario

To further validate the effectiveness and performance advantages of the proposed method, this paper designs a test under heterogeneous starting point conditions: the four USVs in the cluster are deployed at four different spatial locations: (111.2, 212.2), (141.1, 292.2), (212.5, 251.2), and (181.1, 184.1). The test data, as shown in Fig. 11 and Table 3, demonstrate that compared to the NEMP method, the CEMP method achieves a 28.90% optimization in energy consumption. Furthermore, the proposed CERC method in this paper further enhances energy-saving performance, reducing energy consumption by 38.47% compared to NEMP. This experimental data strongly confirms the effectiveness of the constructed energy consumption model and verifies the superior performance of the MLSRC-MAPPO algorithm in complex scenarios.

4.6. Comparative experiments

To further validate the superiority of the method proposed in this chapter over existing cluster path planning methods, this paper conducts comparative experiments based on the QMIX (Davydov et al., 2021), MADDPG (Wang et al., 2022b), and MATD3 (Guo et al., 2024) algorithms. The experimental conditions and environmental parameters are largely consistent with those of the CERC method. After training the agents, they are tested. During the testing process, all USVs in the cluster start from the coordinate (400, 0.0). The precise data of the test results for each comparative method are shown in Table 4, and the planned trajectories of each comparative method are illustrated in Fig. 12.

Table 3
Detailed planning results of different methods under the different starting point.

Method	No./SP	EC/KWh	ST/h	SD/nm	AS/kn	AA
NEMP	1 (112.2, 212.2)	243.68	6.33	220.13	34.76	0.08421
	2 (141.1, 292.2)	427.82	16.33	380.93	23.32	0.01429
	3 (212.5, 251.2)	461.95	11.67	399.20	34.22	0.06286
	4 (181.1, 184.1)	222.91	6.00	208.80	34.80	0.05000
	overall/Average	1356.36	10.08	1209.06	31.77	0.05284
CEMP	1 (112.2, 212.2)	127.09	15.67	213.00	13.60	0.02553
	2 (141.1, 292.2)	404.87	14.00	383.13	27.37	0.00952
	3 (212.5, 251.2)	267.37	28.67	415.40	14.49	0.03140
	4 (181.1, 184.1)	165.06	13.33	205.13	15.39	0.14000
	overall/Average	964.39	17.92	1216.67	17.71	0.05161
CERC	1 (112.2, 212.2)	161.51	13.67	216.20	15.82	0.01220
	2 (141.1, 292.2)	240.72	30.33	369.00	12.16	0.06044
	3 (212.5, 251.2)	274.85	27.33	411.40	15.05	0.01098
	4 (181.1, 184.1)	157.51	13.67	216.00	15.80	0.12439
	overall/Average	834.59	21.25	1212.60	14.71	0.05200

Table 4
Precise planning results of each comparative method.

Method	No./SP	EC/KWh	ST/h	SD/nm	AS/kn	AA
QMIX	1 (400, 0)	545.24	17.67	542.67	30.72	0.00943
	2 (400, 0)	664.29	17.67	594.67	33.66	0.10189
	3 (400, 0)	621.89	19.67	588.67	29.93	0.10508
	4 (400, 0)	499.33	17.33	507.40	28.80	0.09038
	overall/Average	2338.84	18.08	2225.33	30.78	0.07669
MADDPG	1 (400, 0)	531.49	15.00	520.00	34.67	−0.00444
	2 (400, 0)	639.39	16.00	547.20	34.20	0.13958
	3 (400, 0)	628.74	18.33	572.53	31.22	0.09091
	4 (400, 0)	474.12	13.67	433.87	31.74	0.06341
	overall/Average	2273.75	15.75	2073.60	32.96	0.07237
MATD3	1 (400, 0)	505.48	16.67	508.27	30.49	0.01200
	2 (400, 0)	650.68	15.67	537.87	34.33	0.18511
	3 (400, 0)	559.72	19.33	536.53	27.75	0.01724
	4 (400, 0)	460.00	15.00	435.20	29.01	0.07556
	overall/Average	2175.88	16.67	2017.87	30.70	0.07248

The comparison results based on Table 4 and Fig. 12 show that the proposed method significantly outperforms the comparative methods in comprehensive path planning performance metrics, demonstrating outstanding energy efficiency advantages with the lowest total energy consumption. Compared with QMIX, MADDPG and MATD3 methods, it reduces the total energy consumption by 20.83%, 18.56% and 14.9% respectively. This comparison not only validates the technical advancement of the proposed method in multi-objective collaborative optimization but also reveals its engineering value for achieving green navigation in complex marine environments through specific data.

Furthermore, for more comprehensive comparison, this paper tested traditional methods including the Genetic Algorithm (Vicmudo et al., 2014), Particle Swarm Optimization (Lu et al., 2022), and A* algorithm (Wang, 2021). The experimental verification reveals that conventional methods prove inadequate for complex multi-USV scenarios requiring concurrent optimization of heading and speed across four USVs, while simultaneously addressing energy efficiency, dynamic obstacle avoidance, target-reaching objectives, and path conflict resolution. These traditional methods demonstrate prohibitively long optimization durations and convergence failure, ultimately failing to deliver viable energy-efficient swarm path planning solutions. This comparative result strongly corroborates the significant advantages of the proposed method in handling highly complex swarm path planning problems.

4.7. Experimental results under different USV cluster scales

To verify the scalability of the proposed method for multi-USV systems, this paper further conducted simulation experiments with 6 and 8 USVs. The test results are presented in Tables 5, 6 and Fig. 13, respectively. The experimental results demonstrate that in both 6-USV and 8-USV cluster scales, the proposed method effectively achieves energy-efficient cooperative path planning, confirming its strong scalability in multi-agent scenarios.

Table 5
Accurate planning results under 6 USVs.

Method	No./SP	EC/KWh	ST/h	SD/nm	AS/kn	AA
CERC (Proposed)	1 (112.2, 212.2)	141.12	13.67	220.60	16.14	0.00976
	2 (141.1, 292.2)	257.67	27.00	373.40	13.83	0.02469
	3 (212.5, 251.2)	388.59	14.33	403.00	:28.12	0.04186
	4 (181.1, 184.1)	210.44	6.67	204.27	30.64	0.06000
	5 (241.1, 154.2)	254.94	27.00	355.40	13.16	0.02593
	6 (312.5, 201.2)	223.24	22.33	325.20	14.56	0.10597
	overall/Average	1475.99	18.50	1881.87	19.41	0.04470

Table 6
Accurate planning results under 8 USVs.

Method	No./SP	EC/KWh	ST/h	SD/nm	AS/kn	AA
CERC (Proposed)	1 (112.2, 212.2)	160.68	14.00	224.20	16.01	0.00476
	2 (141.1, 292.2)	309.05	29.00	400.20	13.80	0.00575
	3 (212.5, 251.2)	336.37	27.00	444.53	16.46	0.01852
	4 (181.1, 184.1)	242.47	13.67	183.94	17.74	0.04390
	5 (241.1, 154.2)	253.60	24.67	328.80	13.33	0.07027
	6 (312.5, 201.2)	269.65	30.33	383.60	12.65	−0.00769
	7 (201.5, 162.0)	288.52	15.67	338.27	21.59	0.07021
	8 (151.1, 191.2)	223.04	21.33	292.40	13.70	−0.00313
	overall/Average	2024.86	21.96	2654.47	15.66	0.02532

4.8. Impact of action space size on method performance

To investigate the impact of action space dimensionality on method performance, this study sets both the heading and speed adjustment subspace dimensions to 5 to examine their influence on path planning effectiveness (results shown in Fig. 14).

The results in Fig. 14 reveal that when the subspace dimensionality is reduced to 5, the planned trajectory exhibits increased tortuosity and significantly reduced smoothness. This indicates that subspace dimensionality has a substantial impact on trajectory quality, and thus, higher dimensions should be prioritized where possible. However, excessively high dimensionality may lead to convergence difficulties in multi-agent systems. Therefore, an appropriate subspace dimensionality should be selected by balancing task requirements and hardware constraints.

5. Conclusion

Aiming at the problem of collaborative energy-efficient path planning for USV cluster, this paper proposes a systematic solution: based on the theoretical framework of multi-agent reinforcement learning, a cluster reinforcement learning training environment is established by constructing a trinity modeling framework consisting of the USV kinematics model, energy consumption model, and cluster conflict model. Subsequently, an innovative multi-dimensional action space and the MLSRC-MAPPO algorithm are proposed, which significantly improve the convergence efficiency and energy-saving optimization performance.

The training results show that the proposed multi-dimensional action space method achieves remarkable success in lightweighting the action network: the number of parameters is reduced by 89.10%, and the computational complexity is decreased by 88.94%, greatly enhancing training efficiency. Meanwhile, the MLSRC-MAPPO algorithm, by introducing a multi-head attention mechanism, significantly improves the convergence performance of the multi-agent system, fully validating the superiority of this method in enhancing cluster collaborative performance.

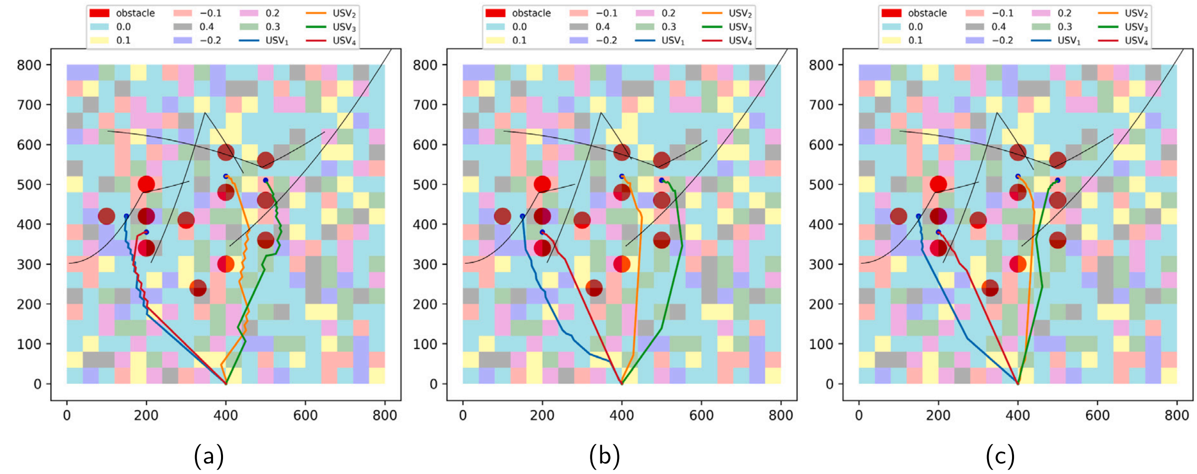

Fig. 12. Path planning trajectories of various comparison methods, (a) QMIX (b) MADDPG (c) MATD3.

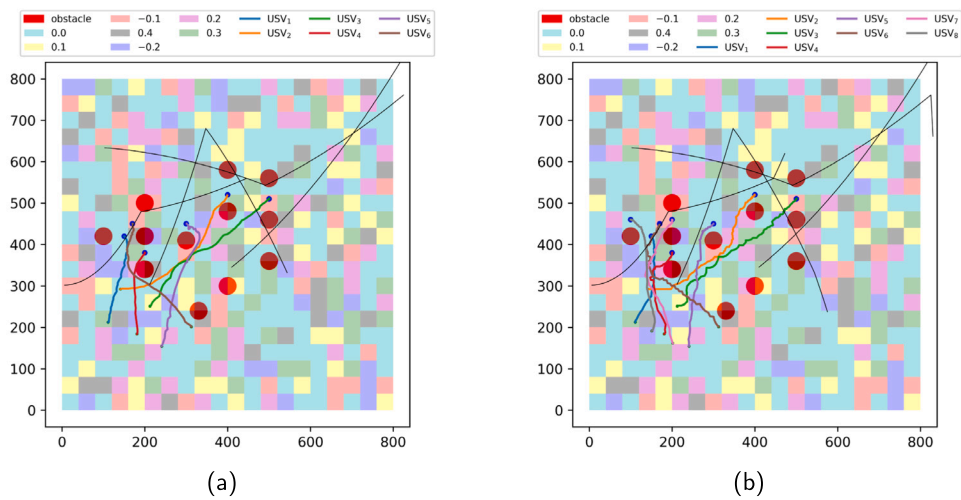

Fig. 13. Planned trajectory, (a) 6 USVs (b) 8 USVs.

The test results indicate that the proposed collaborative energy-efficient path planning method successfully achieves multi-objective collaborative planning for the USV cluster, with all agents safely reaching the target area within the specified time threshold. Through dynamic speed control strategies and a low-energy consumption region optimization mechanism, a significant reduction in energy consumption is achieved. Comparative data from the training phase show that the CERC method significantly outperforms the CEMP method in convergence performance, highlighting the technical advantages of the MLSRC-MAPPO algorithm in enhancing agent collaborative decision-making through the multi-head attention mechanism. For the dual test scenarios of cluster homogeneous/heterogeneous starting points: in the homogeneous starting point scenario, the CEMP method reduces total energy consumption by 15.03% compared to the NEMP baseline. In the heterogeneous starting point scenario, the energy consumption reduction of the CEMP method increases to 28.90%. These test results fully confirm the effectiveness of the established energy consumption model. Notably, the CERC method demonstrates significant energy conservation performance, achieving reductions of 26.24% and 38.47% in the respective scenarios. This breakthrough originates from the MLSRC-MAPPO algorithm's spatiotemporal correlation modeling capability, which establishes dynamic inter-agent coupling relationships to attain Pareto optimality across path length, navigation efficiency, and energy consumption metrics. The comparative experimental results demonstrate that the proposed method comprehensively outperforms existing approaches across all core metrics (navigation speed, operational area, and energy consumption), with its energy efficiency advantage being particularly prominent. Specifically, it achieves total energy consumption reductions of 20.83%, 18.56%, and 14.9% compared to QMIX, MADDPG, and MATD3 methods respectively. These results fully validate both the engineering applicability and academic value of the proposed method in complex marine environments. Furthermore, we verified the good scalability of the proposed method in multi-agent systems by testing scenarios with 6 USVs and 8 USVs respectively.

However, this study also has certain limitations: First, the model validation is currently primarily based on simulations; the robustness of its conclusions under real-world complex and variable sea conditions (such as strong winds/waves, strong currents, or sudden obstacles) requires further field testing and verification. Second, the algorithm's scalability and generality when handling cooperative planning for ultra-large-scale clusters (e.g., dozens or more vessels) or heterogeneous USVs (different hull types, dynamic characteristics) still need in-depth investigation. Furthermore, although training efficiency has been significantly improved, the reinforcement learning training process itself still involves considerable computational resource consumption, posing challenges for real-time deployment on resource-constrained edge computing platforms.

In the future, research efforts can be deepened in the following directions: First, advance the integration of the algorithm into real USV cluster platforms and conduct sea trials, focusing on evaluating

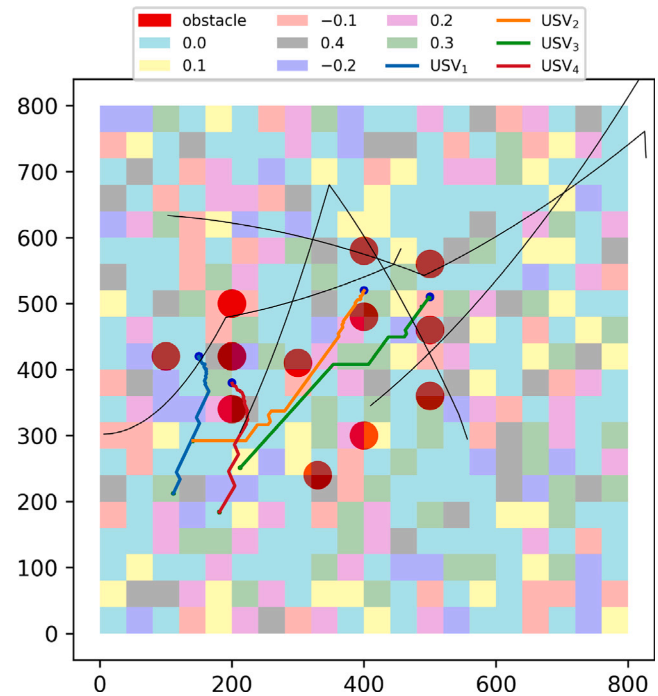

Fig. 14. The planned trajectory under subspace dimensionality of 5.

its actual performance in dynamic, uncertain marine environments and strategies for enhancing robustness. Second, explore scalability optimization of the algorithm, researching adaptive cooperative planning mechanisms for large-scale heterogeneous clusters and dynamic task allocation. Third, investigate lightweight model deployment techniques (such as model pruning, quantization) and distributed training frameworks to reduce computational overhead and enhance practicality in edge computing scenarios. Fourth, consider incorporating more sophisticated environmental perception modules (e.g., fusing real-time oceanographic/meteorological and hydrological data) to develop more intelligent energy-saving strategies adaptable to the environment. Finally, explore integrating the proposed framework with digital twin technology to build a virtual-physical interactive USV cluster cooperative planning validation and optimization platform, providing stronger technical support for the future clustered application of intelligent marine equipment.

CRediT authorship contribution statement

Haipeng Xiao: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lijun Fu:** Writing – review & editing, Supervision, Data curation, Conceptualization. **Chengya Shang:** Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Yunfeng Lin:** Validation, Project administration, Investigation, Funding acquisition, Data curation. **Longfei Yue:** Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Data curation. **Yaxiang Fan:** Visualization, Project administration, Funding acquisition.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

Data will be made available on request.

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